

Aryan Prajapati

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SUMMARY

Final-year CS student (GNDEC; IIT Madras Data Science) who builds **Python** applications with **FastAPI REST APIs**, **SQL databases** and **AI/LLM features**: a Gemini-powered platform live on AWS, a RAG advisor, an audio AI pipeline and ML research on a ~300K-sample dataset.

TECHNICAL SKILLS

Python & Foundations: Python, SQL, C++, TypeScript (basics), Data Structures & Algorithms (50+ LeetCode), Git, Linux, REST APIs, FastAPI

ML & Deep Learning: Pandas, NumPy, Matplotlib, scikit-learn, TensorFlow, PyTorch, Hugging Face, EfficientNet fine-tuning, model training and evaluation (RMSE, ATE), label-noise audits

GenAI & LLMs: LLM integration (Gemini API, Groq), prompt engineering, RAG pipelines, LangChain, ChromaDB, Whisper, Pyannote, PANN

Databases & Cloud: PostgreSQL, MongoDB, Redis, SQLAlchemy, Alembic, Docker, React, AWS (ECS Fargate, ECR, ALB, RDS, ElastiCache, S3, IAM, Secrets Manager, CloudWatch), Render

EXPERIENCE

AI-ML Research Intern

June 2026 – July 2026

CSIR-CMERI

Ludhiana, Punjab

- Built **IMU+GPS (INS+GPS) sensor-fusion** pipelines for trajectory estimation on a ~**300K-sample** dataset.
- Benchmarked **12 methods** (Kalman filter, LSTM, GRU, Mamba, Transformer, BERT, S4 and others) on **RMSE and ATE**; best model **S4** (RMSE 10, ATE 72.951).
- **Deployed** an interactive trajectory-fusion dashboard on **Render** (live demo).

Data Science Intern

June 2025 – July 2025

Punjab AI Excellence

Ludhiana, Punjab

- Audited **label noise and class imbalance**; retraining on cleaned labels cut misclassification ~**12%** on held-out data.
- Fine-tuned **EfficientNet** (37 classes); deployed it for internal demo use.

PROJECTS

Sarathi: AI Career OS | Python, FastAPI, PostgreSQL, Redis, React, Gemini API, Docker, AWS

- Built and deployed solo (live app): a **FastAPI** backend with JWT auth, **PostgreSQL** models and **Gemini-powered job-description analysis**.
- Built an **idempotent seeding pipeline (20 tables)**; fixed a production **Alembic migration conflict**, no data loss.
- AWS: ECS Fargate, ALB, RDS, ElastiCache, Secrets Manager, least-privilege IAM; removed a NAT Gateway, saving ~\$32/month.

Crop Disease Detection + RAG Advisor | FastAPI, React, ChromaDB, MongoDB Atlas, Groq, TensorFlow

- **RAG chatbot** over agronomy documents (ChromaDB, Groq LLM) with React and MongoDB Atlas.
- **EfficientNetB0** classifier, 8 crops and 37 disease classes: **0.94 test accuracy** (PlantVillage, lab-condition images); beat a from-scratch CNN; served via FastAPI.

Audio Reasoning System | Python, FastAPI, PyTorch (CNN14/PANN), Whisper (Groq), Pyannote, Gemini, SQLite

- Backend that fuses a **CNN14 audio-event classifier** (AudioSet, 527 classes), **Whisper** transcription and **Pyannote** speaker diarization so a **Gemini** agent can answer questions about any audio file.
- Cut follow-up queries from ~**30 s to under 2 s** (pipeline runs once per upload, cached in thread-safe sessions); fused speaker, transcript and event streams into a **SQLite timeline** and added a **speech gate** against hallucinated transcripts.

EDUCATION

Guru Nanak Dev Engineering College (GNDEC)

B.Tech, Computer Science & Engineering (CGPA: 8.5/10)

Ludhiana, Punjab

July 2023 – July 2027

Indian Institute of Technology (IIT), Madras

Diploma in Data Science & Applications — DL & GenAI Specialization

Chennai, India

May 2023 – Sep 2026

CERTIFICATIONS

AWS Academy: Data Engineering | AWS Academy: Cloud Foundations | Advanced Certificate in ML & Data Science (IIT Madras)

LEADERSHIP

Content Head | CSI – GNDEC: Explained technical concepts to **500+ students** via content campaigns.

Organising Member | ICCT-SD 2026: Coordinated logistics with IEEE/CSI and global delegates across multi-track sessions.